

Set Algebra — Formula Sheet + Practice

Universe: Fix a universe U . All complements are taken in U : $A^c = U \setminus A$.

Key proof method: $A = B \iff (\forall x)[x \in A \iff x \in B]$.

Basic Definitions

- Union: $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- Intersection: $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- Difference: $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$
- Complement: $A^c = U \setminus A$
- Symm. difference: $A \Delta B = (A \setminus B) \cup (B \setminus A)$

Algebra Laws (Union/Intersection)

- Commutative:

$$A \cup B = B \cup A,$$

$$A \cap B = B \cap A$$

- Associative:

$$(A \cup B) \cup C = A \cup (B \cup C),$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

- Distributive:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

- Idempotent: $A \cup A = A$, $A \cap A = A$
- Absorption: $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$
- Identity: $A \cup \emptyset = A$, $A \cap U = A$
- Domination: $A \cup U = U$, $A \cap \emptyset = \emptyset$

Complement Laws

- Involution: $(A^c)^c = A$
- $A \cup A^c = U$, $A \cap A^c = \emptyset$
- De Morgan:

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

Difference & Symmetric Difference

- Rewrite: $A \setminus B = A \cap B^c$
- $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$
- $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
- $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$
- $(A \setminus B)^c = A^c \cup B$
- $A \Delta B = (A \cup B) \setminus (A \cap B)$
- $A \Delta A = \emptyset$, $A \Delta \emptyset = A$, $A \Delta U = A^c$

Membership Translation (Logic Dictionary) For any element x :

$$x \in A \cup B \iff (x \in A) \vee (x \in B)$$

$$x \in A \cap B \iff (x \in A) \wedge (x \in B)$$

$$x \in A^c \iff \neg(x \in A)$$

$$x \in A \setminus B \iff (x \in A) \wedge \neg(x \in B)$$

$$x \in A \Delta B \iff \text{“exactly one of } x \in A, x \in B\text{” (XOR)}$$

Power Set Properties (for $\mathcal{P}(A)$)

- $A \subseteq B \iff \mathcal{P}(A) \subseteq \mathcal{P}(B)$.
- If A is finite, then $|\mathcal{P}(A)| = 2^{|A|}$.

Cardinality / counting identities (Finite Sets)

- **Two sets (Inclusion–Exclusion):**

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

- **Disjoint Union Special Case:** If $A \cap B = \emptyset$, then

$$|A \cup B| = |A| + |B|.$$

- **Three sets (Inclusion–Exclusion):**

$$|A \cup B \cup C|$$

$$= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

- **Difference:**

$$|A \setminus B| = |A| - |A \cap B|.$$

- **Complement (Relative to U):**

$$|A^c| = |U| - |A|.$$

- **Symmetric Difference:**

$$|A \Delta B| = |A \setminus B| + |B \setminus A| = |A| + |B| - 2|A \cap B|.$$

Cartesian Product Properties

- **Distributive Over Union/Intersection:**

$$A \times (B \cup C) = (A \times B) \cup (A \times C),$$

$$A \times (B \cap C) = (A \times B) \cap (A \times C).$$

- **Distributive over union/intersection (left factor):**

$$(A \cup B) \times C = (A \times C) \cup (B \times C),$$

$$(A \cap B) \times C = (A \times C) \cap (B \times C).$$

- **Difference:**

$$(A \setminus B) \times C = (A \times C) \setminus (B \times C),$$

$$A \times (B \setminus C) = (A \times B) \setminus (A \times C).$$

- **Empty set:**

$$A \times \emptyset = \emptyset, \quad \emptyset \times A = \emptyset.$$

- **Cardinality (finite sets):**

$$|A \times B| = |A| |B|.$$

1 Practice Identities with Solutions

1. **Identity elements.** Prove $A \cup \emptyset = A$ and $A \cap U = A$.

Solution. For any x :

$$x \in A \cup \emptyset \iff (x \in A) \vee (x \in \emptyset) \iff (x \in A) \vee (\text{false}) \iff x \in A.$$

So $A \cup \emptyset = A$. Similarly,

$$x \in A \cap U \iff (x \in A) \wedge (x \in U) \iff (x \in A) \wedge (\text{true}) \iff x \in A,$$

so $A \cap U = A$.

2. **Difference as intersection with complement.** Prove $A \setminus B = A \cap B^c$.

Solution. For any x :

$$x \in A \setminus B \iff (x \in A) \wedge (x \notin B) \iff (x \in A) \wedge (x \in B^c) \iff x \in A \cap B^c.$$

3. **De Morgan (union).** Prove $(A \cup B)^c = A^c \cap B^c$.

Solution. For any x :

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin (A \cup B) \iff \neg((x \in A) \vee (x \in B)) \\ &\iff (\neg(x \in A)) \wedge (\neg(x \in B)) \iff (x \in A^c) \wedge (x \in B^c) \iff x \in A^c \cap B^c. \end{aligned}$$

4. **Distributive law.** Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution. For any x :

$$\begin{aligned} x \in A \cap (B \cup C) &\iff (x \in A) \wedge ((x \in B) \vee (x \in C)) \\ &\iff ((x \in A) \wedge (x \in B)) \vee ((x \in A) \wedge (x \in C)) \\ &\iff x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

5. **Absorption.** Prove $A \cup (A \cap B) = A$.

Solution. For any x :

$$x \in A \cup (A \cap B) \iff (x \in A) \vee ((x \in A) \wedge (x \in B)).$$

In logic, $P \vee (P \wedge Q) \iff P$. Taking $P = (x \in A)$ gives $x \in A \cup (A \cap B) \iff x \in A$.

6. **Difference of a union.** Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Solution. For any x :

$$\begin{aligned} x \in A \setminus (B \cup C) &\iff (x \in A) \wedge \neg((x \in B) \vee (x \in C)) \\ &\iff (x \in A) \wedge (\neg(x \in B) \wedge \neg(x \in C)) \\ &\iff ((x \in A) \wedge \neg(x \in B)) \wedge ((x \in A) \wedge \neg(x \in C)) \\ &\iff x \in (A \setminus B) \cap (A \setminus C). \end{aligned}$$

7. **Complement of a difference.** Prove $(A \setminus B)^c = A^c \cup B$.

Solution. Using $A \setminus B = A \cap B^c$:

$$(A \setminus B)^c = (A \cap B^c)^c.$$

By De Morgan (intersection),

$$(A \cap B^c)^c = A^c \cup (B^c)^c = A^c \cup B.$$

8. **Subtracting twice.** Prove $(A \setminus B) \setminus C = A \setminus (B \cup C)$.

Solution. For any x :

$$\begin{aligned} x \in (A \setminus B) \setminus C &\iff x \in A \setminus B \text{ and } x \notin C \\ &\iff ((x \in A) \wedge (x \notin B)) \wedge (x \notin C) \iff (x \in A) \wedge ((x \notin B) \wedge (x \notin C)) \\ &\iff (x \in A) \wedge \neg((x \in B) \vee (x \in C)) \iff x \in A \setminus (B \cup C). \end{aligned}$$

9. **Symmetric difference identity.** Prove

$$A \triangle B = (A \cup B) \setminus (A \cap B).$$

Solution. Start from the definition:

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

Rewrite differences:

$$(A \setminus B) \cup (B \setminus A) = (A \cap B^c) \cup (B \cap A^c).$$

Now note: “exactly one of A, B ” is the same as “in $A \cup B$ but not in $A \cap B$ ”. Formally, for any x :

$$\begin{aligned} x \in (A \cap B^c) \cup (B \cap A^c) &\iff ((x \in A) \wedge (x \notin B)) \vee ((x \in B) \wedge (x \notin A)) \\ &\iff ((x \in A) \vee (x \in B)) \wedge \neg((x \in A) \wedge (x \in B)) \iff x \in (A \cup B) \setminus (A \cap B). \end{aligned}$$

2 More Set Theory Properties

Standing assumptions: $A, B, C \subseteq U$, complements in U .

A. Subset criteria (very common in proofs)

Claim A1. $A \subseteq B \iff A \cap B = A$.

Proof. ($\Rightarrow$) If $A \subseteq B$, then every $x \in A$ lies in B , so $x \in A \cap B$. Hence $A \subseteq A \cap B$. Also $A \cap B \subseteq A$ always, so $A \cap B = A$.

($\Leftarrow$) If $A \cap B = A$, then $A = A \cap B \subseteq B$, hence $A \subseteq B$. □

Claim A2. $A \subseteq B \iff A \cup B = B$.

Proof. ($\Rightarrow$) If $A \subseteq B$, then $A \cup B \subseteq B$ and also $B \subseteq A \cup B$, so equality holds. ($\Leftarrow$) If $A \cup B = B$, then $A \subseteq A \cup B = B$, so $A \subseteq B$. □

B. More difference identities

Claim B1. $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.

Proof. For any x ,

$$\begin{aligned} x \in (A \cup B) \setminus C &\iff ((x \in A) \vee (x \in B)) \wedge \neg(x \in C) \\ &\iff ((x \in A) \wedge \neg(x \in C)) \vee ((x \in B) \wedge \neg(x \in C)) \iff x \in (A \setminus C) \cup (B \setminus C). \end{aligned}$$

□

Claim B2. $(A \cap B) \setminus C = (A \setminus C) \cap (B \setminus C)$.

Proof. For any x ,

$$\begin{aligned} x \in (A \cap B) \setminus C &\iff (x \in A) \wedge (x \in B) \wedge \neg(x \in C) \\ &\iff ((x \in A) \wedge \neg(x \in C)) \wedge ((x \in B) \wedge \neg(x \in C)) \iff x \in (A \setminus C) \cap (B \setminus C). \end{aligned}$$

□

Claim B3. $A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C)$.

Proof. Start with membership:

$$x \in A \setminus (B \setminus C) \iff (x \in A) \wedge \neg(x \in B \setminus C).$$

But $x \in B \setminus C \iff (x \in B) \wedge \neg(x \in C)$, so

$$\neg((x \in B) \wedge \neg(x \in C)) \iff \neg(x \in B) \vee (x \in C).$$

Hence

$$\begin{aligned} x \in A \setminus (B \setminus C) &\iff (x \in A) \wedge (\neg(x \in B) \vee (x \in C)) \\ &\iff ((x \in A) \wedge \neg(x \in B)) \vee ((x \in A) \wedge (x \in C)) \iff x \in (A \setminus B) \cup (A \cap C). \end{aligned}$$

□

C. Partition-style identities (very useful for counting)

Claim C1. $A = (A \setminus B) \dot{\cup} (A \cap B)$. (Disjoint union.)

Proof. First, $(A \setminus B) \cap (A \cap B) = \emptyset$ because x cannot be simultaneously in B and not in B . Next, for any $x \in A$, either $x \in B$ or $x \notin B$; in the first case $x \in A \cap B$, in the second $x \in A \setminus B$. So $A \subseteq (A \setminus B) \cup (A \cap B)$. The reverse inclusion is clear. □

Corollary (finite sets). If A is finite, then $|A| = |A \setminus B| + |A \cap B|$. □

D. More symmetric-difference properties

Claim D1. $(A \Delta B)^c = (A \cap B) \cup (A^c \cap B^c)$.

Proof. Using $A \Delta B = (A \cap B^c) \cup (B \cap A^c)$,

$$(A \Delta B)^c = (A \cap B^c) \cup (B \cap A^c)^c = (A \cap B^c)^c \cap (B \cap A^c)^c.$$

By De Morgan,

$$(A \cap B^c)^c = A^c \cup B, \quad (B \cap A^c)^c = B^c \cup A.$$

So

$$(A \Delta B)^c = (A^c \cup B) \cap (B^c \cup A).$$

Distribute:

$$(A^c \cup B) \cap (B^c \cup A) = (A^c \cap B^c) \cup (A^c \cap A) \cup (B \cap B^c) \cup (B \cap A).$$

Middle two terms are $\emptyset$, leaving

$$(A \Delta B)^c = (A^c \cap B^c) \cup (A \cap B).$$

□

Claim D2. $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$.

Proof. Start from $B \Delta C = (B \setminus C) \cup (C \setminus B)$:

$$A \cap (B \Delta C) = A \cap (B \setminus C) \cup A \cap (C \setminus B).$$

Now $A \cap (B \setminus C) = A \cap (B \cap C^c) = (A \cap B) \cap (A \cap C)^c = (A \cap B) \setminus (A \cap C)$. Similarly $A \cap (C \setminus B) = (A \cap C) \setminus (A \cap B)$.
Therefore

$$A \cap (B \Delta C) = ((A \cap B) \setminus (A \cap C)) \cup ((A \cap C) \setminus (A \cap B)) = (A \cap B) \Delta (A \cap C).$$

□